Lab Exercise 2: Working with Docker Volumes

Objective:

- Learn how to create and manage Docker volumes.
- Understand how Docker volumes can be used to persist data across container restarts.
- Practice mounting Docker volumes to containers.

Prerequisites:

- Docker installed on your system.
- Basic understanding of Docker commands and container concepts.

Step 1: Create a Docker Volume

Create a new Docker volume:

```
docker volume create my_data_volume
```

This command creates a Docker volume named my_data_volume.

Verify that the volume was created:

```
docker volume ls
```

You should see my_data_volume listed among the volumes.

```
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>docker volume create my_data_volume
my_data_volume

C:\Users\NK>docker volume ls
DRIVER    VOLUME NAME
local     my_data_volume

C:\Users\NK>
```

Step 2: Run a Container with the Volume Mounted

Run an Nginx container with the volume mounted:

```
docker run -d --name my_nginx -v my_data_volume:/usr/share/nginx/html -p 8008:80 nginx
```

This command starts an Nginx container named my_nginx and mounts the my_data_volume volume to the /usr/share/nginx/html directory inside the container.

Verify that the container is running:

```
docker ps
```

You should see my_nginx listed as one of the running containers.

```
C:\Users\NK>docker volume ls
DRIVER    VOLUME NAME

C:\Users\NK>docker run -d --name my_nginx -v my_data_volume:/usr/share/nginx/html -p 8009:80 nginx
f56d7f80013ff558c336e6d01ad9189af286fde412dc8e22e169ace8127875fd

C:\Users\NK>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                                       NAMES
f56d7f80013f   nginx     "/docker-entrypoint.…"   13 seconds ago   Up 12 seconds   0.0.0.0:8009->80/tcp, [::]:8009->80/tcp   my_nginx

C:\Users\NK>
```

Step 3: Interact with the Volume

Create a simple HTML file in the volume:

```
docker exec -it my_nginx bash

echo "<h1>Hello, Docker Volume</h1>" > /usr/share/nginx/html/index.html

exit
```

This command creates an HTML file inside the /usr/share/nginx/html directory, which is backed by my_data_volume.

Access the Nginx server to see your file: Open a browser and navigate to http://localhost:8008. You should see the message "Hello, Docker Volume!" displayed on the page.

```
C:\Users\NK>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\NK>docker exec -it my_nginx bash
root@f56d7f80013f:/# echo "<h1>Hello, Docker Volume</h1>" > /usr/share/nginx/html/index.html
root@f56d7f80013f:/# exit

C:\Users\NK>
```

Step 4: Test Data Persistence

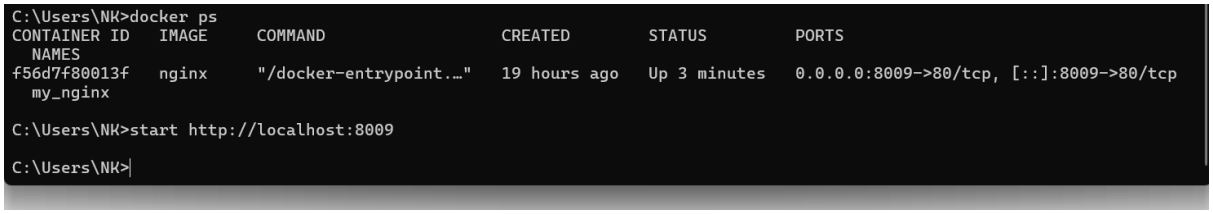
Stop and remove the container:

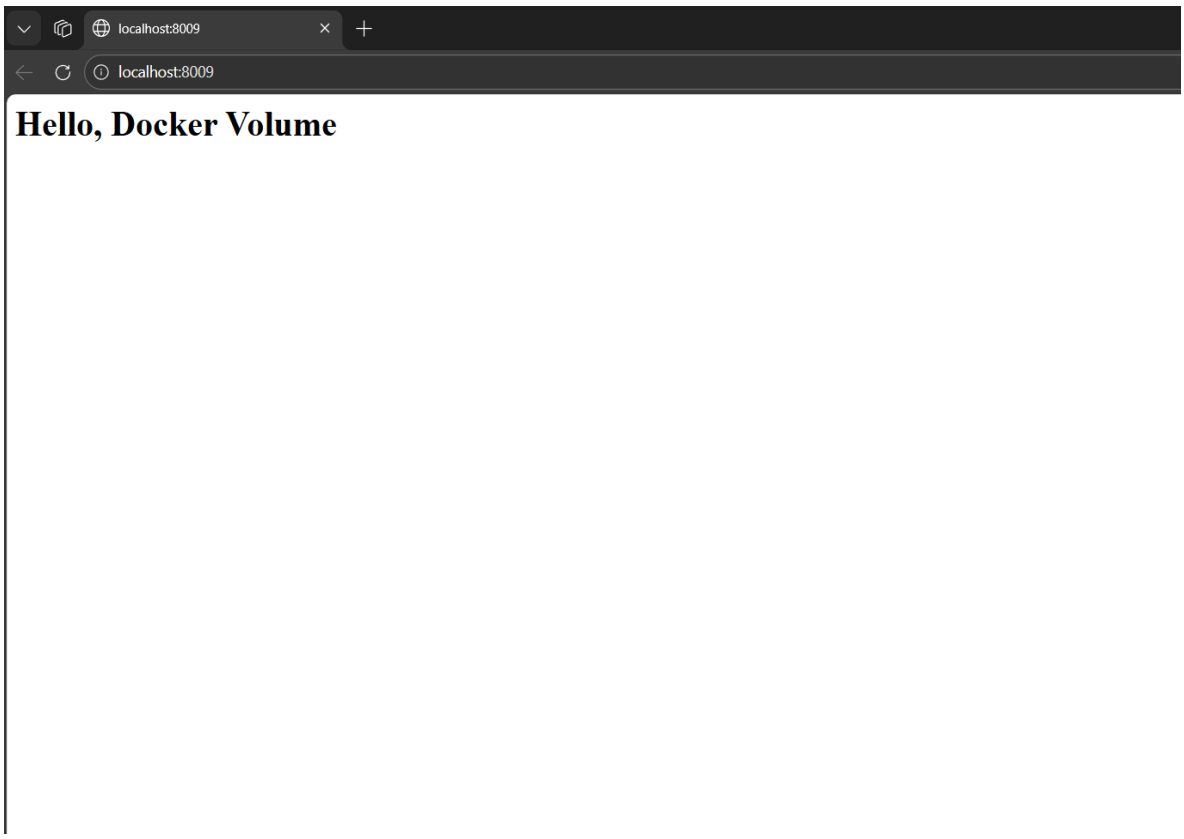
```
docker stop my_nginx

docker rm my_nginx
```

Run a new Nginx container using the same volume:

```
docker run -d --name my_nginx -v my_data_volume:/usr/share/nginx/html -p 8008:80 nginx
```

Access the Nginx server again: Navigate to http://localhost in your browser. You should still see the "Hello, Docker Volume!" message, demonstrating that the data persisted across container instances.





Step 5: Clean Up

Stop and remove the container:

```
docker stop my_nginx

docker rm my_nginx
```

Remove the Docker volume:

```
docker volume rm my_data_volume
```

Verify that the volume is removed:

```
docker volume ls
```

Ensure that my_data_volume is no longer listed.

```
C:\Users\NK>start http://localhost:8009

C:\Users\NK>docker stop my_nginx
my_nginx

C:\Users\NK>docker rm my_nginx
my_nginx

C:\Users\NK>docker run -d --name my_nginx -v my_data_volume:/usr/share/nginx/html -p 8008:80 nginx
b43f2625e29d771dbf733dc5e37aa581226927593e01c263b1c923e5199d922e

C:\Users\NK>http://localhost:8008
'http:' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\NK>start http://localhost:8008

C:\Users\NK>docker stop my_nginx
my_nginx

C:\Users\NK>docker rm my_nginx
my_nginx

C:\Users\NK>docker volume rm my_data_volume
my_data_volume

C:\Users\NK>docker volume ls
DRIVER    VOLUME NAME
```

```
C:\Users\NK>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS         PORTS
  NAMES
f56d7f80013f   nginx     "/docker-entrypoint.…"   19 hours ago   Up 3 minutes   0.0.0.0:8009->80/tcp, [::]:8009->80/tcp
  my_nginx

C:\Users\NK>start http://localhost:8009

C:\Users\NK>docker stop my_nginx
my_nginx

C:\Users\NK>docker rm my_nginx
my_nginx

C:\Users\NK>docker run -d --name my_nginx -v my_data_volume:/usr/share/nginx/html -p 8008:80 nginx
b43f2625e29d771dbf733dc5e37aa581226927593e01c263b1c923e5199d922e

C:\Users\NK>http://localhost:8008
'http:' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\NK>start http://localhost:8008

C:\Users\NK>
```